

Homework 3 - 193DS Spring

Michael Torbensen

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```
# attaching packages

library(tidyverse)
library(janitor)
library(here)
library(effectsize)
library(readxl)
library(ggeffects)
library(rstatix)

# reading in kelp data
kelp <- read.csv(
  here("data", "temp_kelp.csv"))

# reading in personal data
my_data <- read.csv(
  here("data", "Personal_data.csv"))
```

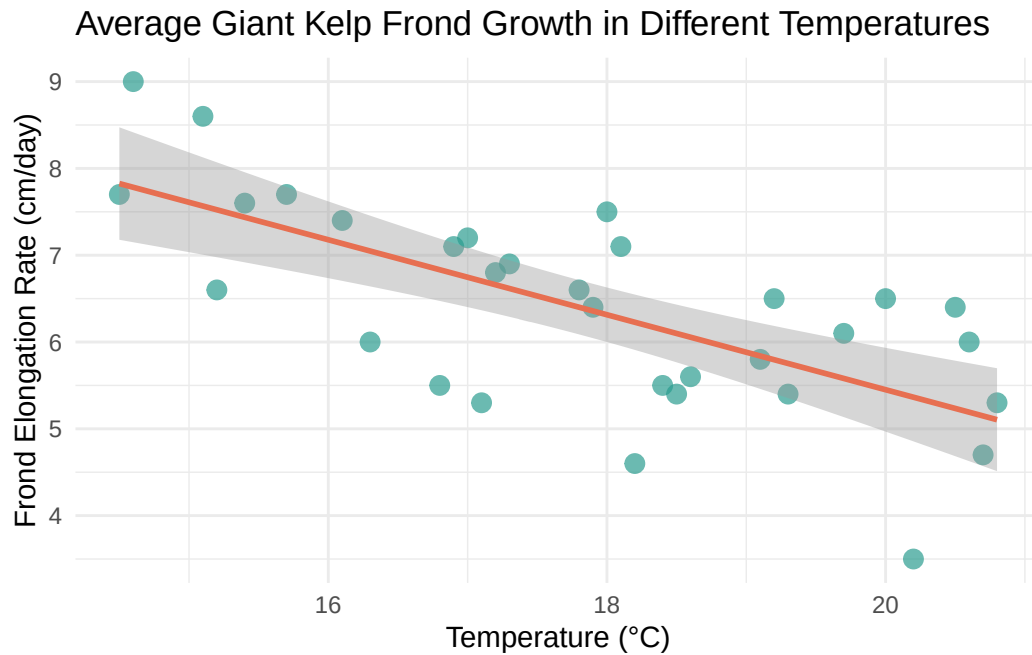
Problem 1. Giant Kelp Fronds

a. An appropriate test

To start off, in order to measure the strength and direction of the relationship between kelp length and temperature I will use Pearson's Correlation Coefficient. In order to visualize and model the relationship, I will use a simple linear regression. The Pearson's Correlation Coefficient provides a numerical value to describe the relationship between the two variables and the simple linear regression models this relationship with a line.

b. Create a visualization

```
# base layer: ggplot
ggplot(data = kelp,
       # x-axis: temperature
       # y-axis: kelp elongation rate (cm/day)
       mapping = aes(x = temp_c,
                     y = kelp_elong)) +
# first layer: adding data points
geom_point(color = "#2a9d8f",
           alpha = 0.7,
           size = 3) +
# second layer: adding trend line with color and standard error
geom_smooth(method = "lm",
            color = "#e76f51",
            se = TRUE) +
# labeling x- and y-axis
labs(
  x = "Temperature (°C)",
  y = "Frond Elongation Rate (cm/day)",
  title = "Average Giant Kelp Frond Growth in Different Temperatures") +
# adding different theme from default
theme_minimal()
```

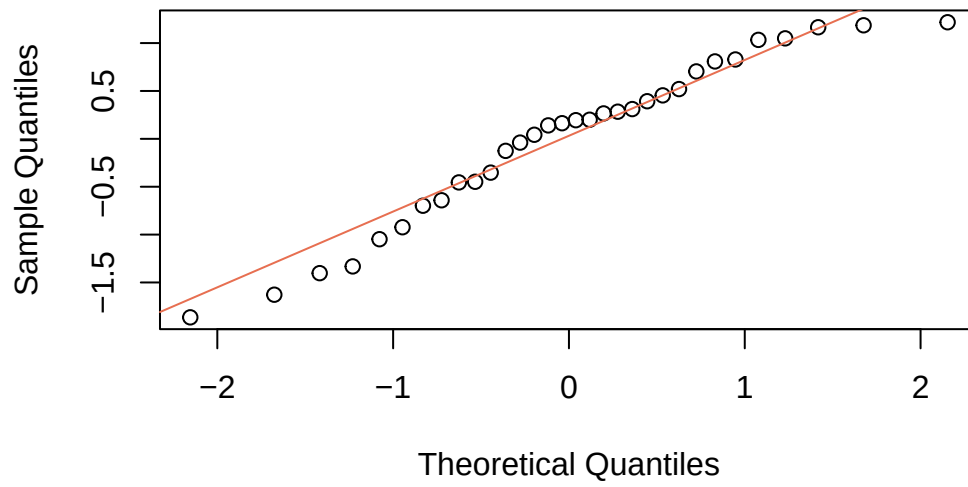


c.

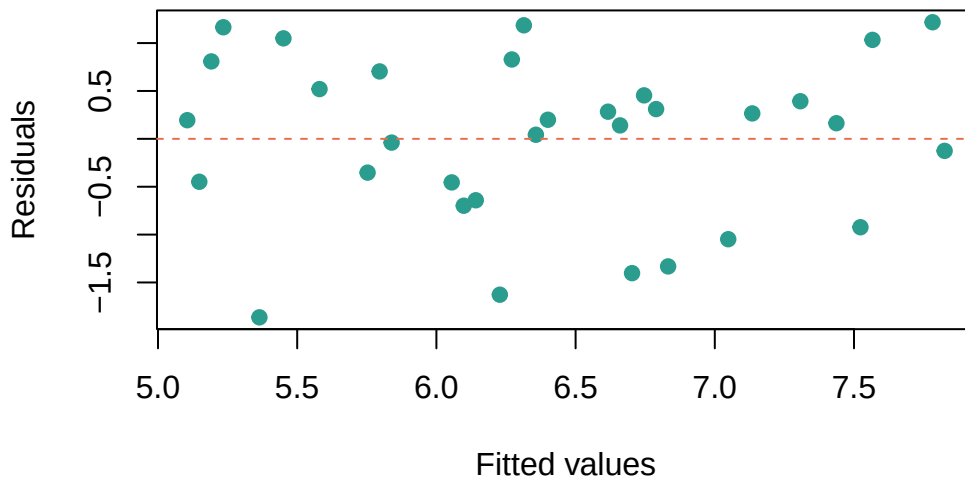
```
# Fit the model first to extract residuals ----
kelp_model <- lm(
  kelp_elong ~ temp_c,
  data = kelp)

# Check normality of residuals: Q-Q plot ----
qqnorm(residuals(kelp_model))
qqline(residuals(kelp_model),
  col = "#e76f51")
```

Normal Q-Q Plot



```
# Check homoscedasticity: residuals vs fitted values ----
plot(fitted(kelp_model),
     residuals(kelp_model),
     xlab = "Fitted values",
     ylab = "Residuals",
     pch = 19,
     col = "#2a9d8f")
abline(h = 0,
       col = "#e76f51",
       lty = 2)
```



```
# Pearson's correlation coefficient test ----
cor.test(kelp$temp_c,
         kelp$kelp_elong,
         method = "pearson")
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

```
# Simple linear regression ----
summary(kelp_model)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-1.8643	-0.5017	0.1790	0.5659	1.2177

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.08645	1.49219	9.440	1.72e-10 ***
temp_c	-0.43179	0.08321	-5.189	1.37e-05 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.8665 on 30 degrees of freedom
```

```
Multiple R-squared:  0.473, Adjusted R-squared:  0.4554
```

```
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

Linearity was assessed visually using a scatterplot of temperature vs. elongation rate with a fitted regression line. Normality of residuals was checked using a Q-Q plot, and homoscedasticity was assessed using a residuals vs. fitted values plot. The scatterplot suggests a reasonably linear relationship, the Q-Q plot shows residuals falling close to the reference line, and the residuals vs. fitted plot shows no obvious pattern. All three assumptions appear reasonably satisfied too.